

QFin Project Overview -- Data

Alright let's just get into it.

You will be trading various different products on a mock exchange. You will create a bot to place orders on these products both against each other and against bots.

You will have three weeks to do this.

Products

Fruit

The base products you will be trading are five fruits. Apples, oranges, bananas, strawberries, and watermelon. Creative I know.

Each fruit has a fair value. However, the price of the fruit is not going to match the fair value exactly. The price is dictated by order flow. There is a market making bot that is uninformed that will adjust its prices based on the trades made against it. Currently there are both uninformed bots providing flow and informed bots that push the price in the direction of the fair value.

The fruits all move according to some sort of process. Your general goal is to figure out mispricings in the market. There are a lot of patterns in how the prices move by design. Obviously if you figure these out you will profit by being able to move the market in the direction you choose.

ETF

There is an 'ETF' that you can trade. A basket of fruit. So creative. The fair price of the ETF is literally the sum of all the fruit. It is nothing more complicated than that. Again there is a market maker bot, uninformed flow, and informed bots trading the ETF in the right direction.

When we get to using the actual exchange, you will be allowed to create and redeem ETFs for a fee. What is the fee? TBD.

Up-down Options

Similar to Polymarket BTC up down options, you will have the ability to trade 5 minute and 60 minute up-down options on the fruit ETF. Currently the data has 15 minute up-down options as well. I have thought about this and it seems a bit ludicrous to give you 3 up-down options to trade. So, I will likely not keep this in the final exchange.

Up-down options are priced as a percentage. So, if you buy the option for 60 cents, that is equivalent to believing the ETF has a 60% chance of going up over the period.

The strike is the price of the last ETF trade at or before the open second. The settle is the last ETF trade at or before the close second. If the ETF didn't trade in that second, it's the most recent earlier trade. Do I think that this will create issues? Yes, but alas, we shall see how it goes.

The 15 minute up down option actually settles differently and is based on the mid price at the strike second and settlement section. I did this because on Polymarket the 5 minute, 15 minute, and 4 hour markets settle differently to the 1 hour market. Which is so stupid that I think it needed to be incorporated here.

If the settlement price is equal exactly to the strike price, the option resolves to up.

The Data

You can download the data from <https://drive.google.com/file/d/1pQ1dnIGD9wdyja-kUKqnb7I2suFFtxbf/view?usp=sharing>. It is large.

Each section below is a folder of Parquet files in the data. There is one file per UTC day.

```
/<table>/date=YYYY-MM-DD.parquet
```

All times are UTC and are seconds after the epoch:

[https://en.wikipedia.org/wiki/Epoch_\(computing\)](https://en.wikipedia.org/wiki/Epoch_(computing)). All prices are in dollars.

Where there is a column named 'product' it is one of `apples`, `bananas`, `watermelon`, `strawberries`, `oranges`, `etf`.

top_of_book

The best bid and ask for each fruit and the ETF, every second.

Column	Meaning
ts	The second. The snapshot is taken at the end of that second.
product	Which product.
bid	Best bid price.
ask	Best ask price.
bid_size	Units available at the best bid.
ask_size	Units available at the best ask.

trades

Every trade in the fruits and the ETF.

Column	Meaning
ts	The second the trade happened. Several trades can share a second. They are listed in order.
product	Which product.
price	Average fill price of the order. It can be off the 1c grid when the order fills across several price levels.
size	Units traded.
aggressor	buy : the buyer crossed the spread. sell : the seller crossed the spread.

option_top_of_book

The best bid and ask for each option expiry, every second.

Each option is an up contract on the ETF. It pays \$1 if the ETF finishes at or above the strike, and \$0 otherwise. A down contract is the other side, so its price is 1 minus the up price.

Column	Meaning
ts	The second.
expiry_min	Window length in minutes: 5, 15 or 60.
window_open	Start time of the window this quote belongs to.
bid	Best bid for the up contract, between 0.01 and 0.99. Empty when that side is not quoted.
ask	Best ask for the up contract, between 0.01 and 0.99. Empty when that side is not quoted.
bid_size	Contracts at the best bid. 0 when not quoted.
ask_size	Contracts at the best ask. 0 when not quoted.

option_trades

Every option trade. All trades are recorded as the up contract. Buying down at price p shows as sell_up at 1 minus p.

Column	Meaning
ts	The second the trade happened.
expiry_min	Window length in minutes: 5, 15 or 60.
window_open	Start time of the window traded.
price_up	Price of the up contract.

Column	Meaning
size	Contracts traded.
aggressor	buy_up : the buyer of up crossed the spread. sell_up : the seller of up crossed the spread.

option_windows

One row per option window, with its strike and result.

Column	Meaning
expiry_min	Window length in minutes: 5, 15 or 60.
window_open	Start time of the window.
strike	ETF price at the window open.
settle	ETF price at the window close.
up_won	True if settle is at or above strike . A tie counts as up.
price_source	Which ETF price sets strike and settle . etf_mid : the ETF mid. etf_last_trade : the last ETF trade at or before that second.

Each window's settle is the next window's strike .

Scoring

I suspect we will open the actual live exchange next week, giving you around two weeks to have access to live trading and get familiar with it.

We will have two phases: the training phase and the testing phase. In training, everyone's PnL's and positions will reset at the end of each day. At the start of the day you will have \$10k to begin with and use to trade. If you run out of the money during a day at all, I will let you just reset to 10k and no position. However, this will likely hurt you in the testing phase so don't just get a high variance strategy expecting that if you win on 20% of days you will get a high score. You will not.

As I am writing this it occurs to me that you will be able to short sell and profit infinitely. We are going to need to look at that. fruits will likely just trade as some sort of future. You aren't just gonna be able to short sell and profit a bunch that's what matters. I will let you know when this is sorted, what margin requirements there will be, etc. (Ask AI what futures are if you are unsure).

During the testing phase, we will simulate your bots running for, say, 6 months. Each day you will stat with \$10k. Then your score will be the average PnL per day minus the standard deviation of your PnLs (The SIG algothon scoring system).

Presentation

You will need to present as a group your findings. We are doing this so that you actually understand what you are doing instead of just getting Claude on it. While I suspect AI will be incredibly helpful, using it at the cost of understanding is not going to help you.

Your presentation and understanding will contribute a multiplier to your score. What the multiplier is, we will figure out as well. More details to come.

Some Notes

- When we are actually trading on the live exchange some of the patterns will fade solely because of all of your bots trading. Your 'alpha' on the price may decay. This is how markets become more efficient
- There is a lot of low-hanging fruit (hahahaha) in how the products move. There are more difficult patterns to find, but I hope you will be able to find something so you gain some sort of momentum.
- Focus much more on researching how the products move, or how to price the derivatives, than profiting initially. The important part of statistical arbitrage is the statistical part. I say this mainly because we have to figure out the quirks of the
- There is a lot of uninformed order flow. You can capitalise off of this by market making. Think about how you might market make best to capture the most amount of edge.
- Your bots speed will be important. If you have a perfectly optimised bot and an unoptimised bot trying to send the same order at the same moment, the optimised bot will win. Ask Josh about IMC ethernet cables in his project at work.
- The most money to be made is undoubtedly in options. It is also the hardest to work on.
- I reserve the right to prevent you from certain activities that just destroy the point of the exchange. I will let most things slide if they aren't ridiculous (e.g., the exchange is closing and we are going to generate a score for you so you buy infinite ETFs and destroy the settlement price. I am including this so no one gets mad if I do need to change something because of certain trading activities.
- Using AI is great but there are some ideas AI just won't come up with in here (I suspect). Use it where you want and to help with coding but don't do it without understanding.